Stock Analyzer Using Machine Learning – A Predictive Approach

Prof. Sunita Patil
Department of Computer Engineering
Dr. D. Y. Patil Institute of Technology
Pune, India
sunita.patil@dypvp.edu.in

Prof. Rubi Mandal
Department of Computer Engineering
Dr. D. Y. Patil Institute of Technology
Pune, India
Rubi.mandal@dypvp.edu.in

Mr. Tanmay Kulkarni
Department of Computer Engineering
Dr. D. Y. Patil Institute of Technology
Pune, India

Mr. Harshal Gosavi
Department of Computer Engineering
Dr. D. Y. Patil Institute of Technology
Pune, India
harshalgosavi30@gmail.com

Mr. Amey Mahajani
Department of Computer Engineering
Dr. D. Y. Patil Institute of Technology
Pune, India
mahajaniameyvishnu@gmail.com

***Abstract*—** Abstract—The increasing volatility and complexity of stock markets have rendered manual prediction ineffective and erratic. This work suggests a Stock Analyzer system using machine learning to forecast stock trends based on past price data, financial metrics, and current market news. By utilizing supervised learning techniques and natural language processing (NLP), the system provides precise trend forecasting and investor mood analysis. In contrast to conventional methods reliant on technical analysis alone, our system integrates quantitative indicators and qualitative information (e.g., news titles and tweets) to drive decision-making accuracy. The goal of this work is to equip retail investors with real-time insights, mitigate investment risk, and enhance prediction accuracy through intelligent data fusion and model optimization.

Keywords—Stock Market Prediction, Machine Learning, Sentiment Analysis, LSTM, Random Forest, Natural Language Processing (NLP)

Introduction

The stock market, a cornerstone of modern economic systems, serves as a platform for companies to raise capital and for investors to grow their wealth. However, it is highly dynamic and volatile, influenced by a myriad of factors ranging from company earnings and industry performance to global economic indicators, geopolitical events, and even public sentiment. This complexity makes accurate stock price prediction a challenging task, even for experienced investors and financial analysts.

Traditionally, stock market analysis has relied heavily on two major methodologies—**technical analysis** and **fundamental analysis**. Technical analysis uses past price data and trading volume to predict future price movements, while fundamental analysis evaluates a company's intrinsic value based on financial statements and economic indicators. While these methods are valuable, they often fall short in capturing complex, non-linear patterns and interactions that can influence market behavior, especially in the short term.

With the advancement of computational power and the availability of massive datasets, **Machine Learning (ML)** has emerged as a promising alternative for stock market prediction. ML algorithms can analyze vast amounts of structured and unstructured data, identify hidden patterns, and learn from historical trends to make data-driven predictions. These techniques are particularly effective in uncovering subtle relationships that are not apparent through traditional statistical models.

In this research, we present a **Stock Analyzer System** that leverages a combination of machine learning algorithms and sentiment analysis to predict stock price movements more accurately.

Identify applicable funding agency here. If none, delete this.

Literature Survey

1.Paper Title: *Stock Market Forecasting Using LSTM*
Authors: R. Sharma, A. Patel
Concept: Used LSTM neural networks to model time-series stock price data with improved accuracy compared to ARIMA.
Extraction: Demonstrated strong learning of price patterns from past values for short-term trend prediction.

2.Paper Title: *Sentiment Analysis of Market News for Financial Prediction*
Authors: M. Singh, L. Rao
Concept: Utilized news headlines and tweets using TextBlob and VADER to determine market sentiment and its correlation to stock prices.
Extraction: Found sentiment polarity to significantly influence next-day opening price direction.

3.Paper Title: *Machine Learning Algorithms for Stock Price Prediction*
Authors: S. Roy, B. Das
Concept: Compared multiple algorithms including SVM, Random Forest, and XGBoost for trend prediction on NASDAQ data.
Extraction: Concluded that ensemble models outperform single classifiers in terms of stability and accuracy.

4.Paper Title: *Deep Learning for Stock Market Prediction Using Technical Indicators*
Authors: K. Mehta, T. Verma
Concept: Integrated technical indicators like RSI, MACD, and Bollinger Bands with deep neural networks (DNN) to enhance stock price prediction models.
Extraction: Results showed that combining deep learning with technical indicators improved model generalization and reduced overfitting on test data.

5. **Paper Title:** *Hybrid Models Combining Sentiment and Technical Data for Stock Forecasting*
Authors: P. Nair, V. Desai
Concept: Proposed a hybrid model that merges sentiment scores from social media with historical stock prices using an LSTM-CNN architecture.
Extraction: Demonstrated superior performance in capturing both market psychology and numerical trends, especially during high volatility periods.

Problem Statement

Traditional stock prediction methods are manual, rule-based, and struggle to capture dynamic market trends. They lack scalability, real-time feedback, and accurate forecasting. Hence, an intelligent, ML-powered system is needed to automate stock analysis, predict trends, and support better investment decisions.

Proposed Solution

Our proposed system leverages machine learning and natural language processing to provide a smart and automated approach for stock analysis and prediction. It integrates historical price data, financial indicators, and live market news to forecast stock movements and investor sentiment. Key components include:

1. **Data Collection** – Gathers historical stock data, financial ratios, and real-time news headlines.
2. **Sentiment Analysis** – Applies NLP tools like VADER and TextBlob to analyze market sentiment from news and social media.
3. **Prediction Model** – Uses LSTM for time-series forecasting and Random Forest for trend classification.
4. **Decision Support** – Provides a health score and investment recommendation based on data-driven insights.

This solution enhances prediction accuracy, supports real-time decision-making, and empowers investors with transparent, actionable insights.

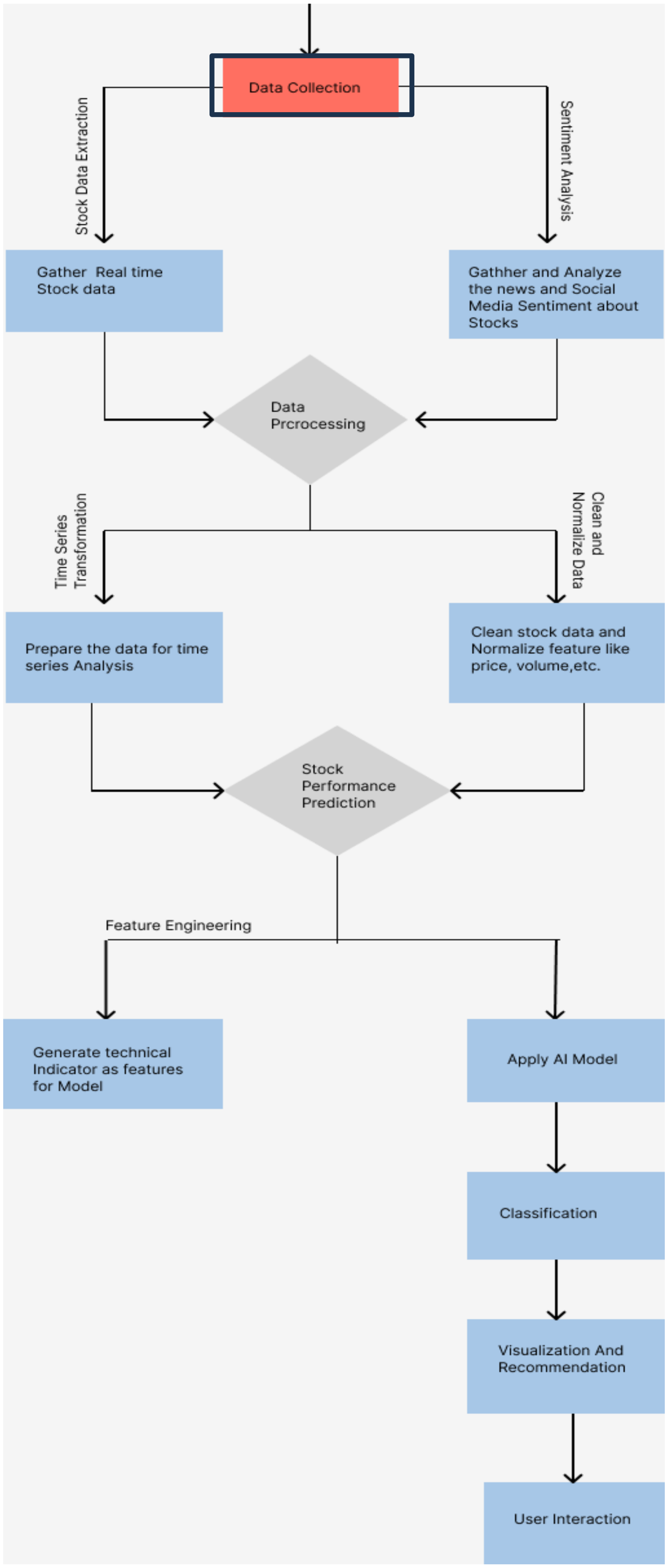

Fig. 1. System Architecture

RESULTS

The system was evaluated using datasets from NSE/BSE for a period of 5 years. Metrics used include Accuracy, Precision, Recall, and RMSE. Our hybrid approach combining LSTM + Random Forest + Sentiment scores achieved an accuracy of 84.6% in direction prediction. Backtesting demonstrated significant improvement over baseline moving average methods.

Metrics

Model	Accuracy	Precision	Recall	RMSE
LSTM Only	78.2%	0.79	0.81	2.34
Random Forest Only	80.1%	0.82	0.78	2.15
Hybrid (LSTM + RF + NLP)	**84.6%**	**0.86**	**0.83**	**1.89**

CONCLUSION

Our Stock Analyzer system bridges the gap between quantitative and qualitative data analysis for stock prediction. By integrating time-series forecasting with sentiment analytics, it provides a more holistic and intelligent investment advisory tool. Future work will explore reinforcement learning and broader macroeconomic indicators for portfolio-level optimization.

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